

Steps to access the Database for this project:

Ssh into the EC2 and log in with the user.

Run the Command:

- `psql -h gradegator-psql.c7kq2myoy2j2.us-east-2.rds.amazonaws.com -U postgres -d postgres -p 5432` (it's the DB-endpoint, Master-username, DB-name)

Password : Bransh05050

This will take you to the psql shell where you can do queries and view the database.

Instructions for accessing the database through a new EC2 server :

Requirements:

- The EC2 instance is in the same VPC as the RDS (or has correct routing)
- EC2 security group is added to the RDS inbound rules on port 5432
- You know the RDS endpoint, Master username, password, and DB name

Dependencies:

- OpenSSH client already available to SSH into the EC2
- psql client installed:
 - Amazon Linux: `sudo yum install postgresql -y`
 - Ubuntu: `sudo apt install postgresql-client -y`

(check by `psql -v` version, if you see the version its installed)

- Terminal command will ask for password → Must use the one from RDS setup
- run the command: `psql -h gradegator-psql.c7kq2myoy2j2.us-east-2.rds.amazonaws.com -U postgres -d postgres -p 5432` (it's the DB-endpoint, Master-username, DB-name)

Password : Bransh05050

This will take you to the psql shell where you can do queries and view the database.

Instructions for accessing the database through a localhost:

Requirements:

- RDS must have Public Access: Enabled
- Your local IP must be added to the RDS Security Group inbound rules (PostgreSQL / 5432)
- You know the RDS endpoint, username, password, and DB name

Dependencies:

In the terminal have the PostgreSQL client installed: (check `psql --version` to see if installed already)

- Mac: `brew install postgresql`
- Ubuntu: `sudo apt install postgresql-client -y`
- Windows: Use [PostgreSQL Installer](#) or use psql via WSL

Run the Command:

- `psql -h gradegator-psql.c7kq2myoy2j2.us-east-2.rds.amazonaws.com -U postgres -d postgres -p 5432` (it's the DB-endpoint, Master-username, DB-name)

Password : Bransh05050

Ensure port 5432 is not blocked by local firewall or network policy

(There might be some different instructions for a local windows powershell)

This will take you to the psql shell where you can do queries and view the database.

A Short Crash Course on PSQL Command and Queries

DATABASE INSPECTION

Task	Command
List databases	<code>\l</code>
Connect to a DB	<code>\c your_db_name</code>
Show current DB	<code>SELECT current_database();</code>
List all tables	<code>\dt</code>
Describe table schema	<code>\d table_name</code>
List users/roles	<code>\du</code>

QUERYING DATA

Task	Command
Show rows from table	<code>SELECT * FROM table_name;</code>
Limit results	<code>SELECT * FROM table_name LIMIT 10;</code>
Filter rows	<code>SELECT * FROM table WHERE column = 'value';</code>
Count rows	<code>SELECT COUNT(*) FROM table_name;</code>

MODIFYING DATA (CRUD)

Action	Command
Insert	<code>INSERT INTO table_name (col1, col2) VALUES ('val1', 'val2');</code>
Update	<code>UPDATE table_name SET col1 = 'new' WHERE id = 1;</code>
Delete	<code>DELETE FROM table_name WHERE id = 1;</code>

SCHEMA CHANGES

Task	Command
Create table	<code>CREATE TABLE table_name (id SERIAL PRIMARY KEY, name TEXT);</code>
Add column	<code>ALTER TABLE table_name ADD COLUMN new_col TEXT;</code>
Remove column	<code>ALTER TABLE table_name DROP COLUMN old_col;</code>
Rename table	<code>ALTER TABLE old_name RENAME TO new_name;</code>
Drop table	<code>DROP TABLE table_name;</code>

TIPS WHILE WORKING

- Always end SQL commands with a ;
- Use arrow keys ↑ ↓ to cycle previous commands
- Use \h to get SQL syntax help
- Use \? to see all `psql` commands

